

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**Deep Learning-Based Brain Tumour Detection Using Convolutional
Neural Networks on MRI Images**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

**Garv Saini
2210990315**

**Arshdeep Singh
2210990171**

Supervised By:

**Dr. Gifty Gupta
Assistant Professor
Department of Computer
Science and Engineering
Chitkara University**



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA**

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **“Deep Learning-Based Brain Tumour Detection Using Convolutional Neural Networks on MRI Images”** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Garv Saini
18 May 2026
2210990315

Arshdeep Singh
18 May 2026
2210990171

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Gifty Gupta

Assistant Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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Garv Saini
2210990315

Arshdeep Singh
2210990171

1. ABSTRACT

Brain tumours are among the most dangerous neurological diseases, and early detection plays a vital role in improving patient survival and treatment planning. Traditional methods of analysing Magnetic Resonance Imaging (MRI) scans mainly rely on manual examination by radiologists, which can be time-consuming and may lead to human errors. With the advancement of Artificial Intelligence and Deep Learning, automated medical image analysis has become more efficient and reliable.

This project presents a deep learning-based approach for detecting brain tumours from MRI images using Convolutional Neural Networks (CNN). The proposed system is designed to classify MRI scans into tumour and non-tumour categories automatically. Various preprocessing techniques such as image resizing, normalization, and data augmentation are applied to improve model performance and reduce overfitting. The CNN model is trained on a publicly available MRI dataset and tested for classification accuracy.

The experimental results demonstrate that the proposed model achieves high accuracy in detecting brain tumours and performs better than many traditional machine learning approaches. The system reduces dependency on manual diagnosis and assists medical professionals in making faster and more accurate decisions. This work highlights the potential of deep learning techniques in the healthcare sector and provides a foundation for future research in multi-class brain tumour classification and real-time clinical applications.

2. INTRODUCTION

Brain tumours are abnormal growths of cells inside the brain that can be either benign (non-cancerous) or malignant (cancerous). These tumors can severely affect important brain functions and may become life-threatening if not detected at an early stage. Accurate and timely diagnosis is essential for proper treatment planning and improving the survival rate of patients. Among different medical imaging techniques, Magnetic Resonance Imaging (MRI) is one of the most commonly used methods for detecting brain abnormalities because it provides high-resolution and detailed images of brain tissues.

Traditionally, radiologists examine MRI scans manually to identify the presence of tumors. However, manual analysis is time-consuming, expensive, and sometimes prone to human errors, especially when dealing with a large number of MRI images. With the advancement of Artificial Intelligence and Deep Learning technologies, automated systems for medical image analysis have gained significant attention. Deep learning models can automatically learn meaningful features from images and perform classification tasks with high accuracy, reducing the dependency on manual diagnosis.

The main objective of this project is to develop an accurate and efficient automated brain tumor detection system that can assist medical professionals in diagnosing tumors more quickly and reliably. The proposed approach aims to reduce human effort, improve classification accuracy, and provide a dependable solution for medical image analysis. Future improvements can include multi-class tumor classification and integration with real-time healthcare applications.

3. METHODOLOGY

The proposed system uses a deep learning-based approach to detect brain tumors from MRI images. The methodology begins with collecting MRI brain scan images from a publicly available dataset containing tumor and non-tumor classes. These images are then prepared for training through preprocessing techniques such as image resizing and normalization. Data augmentation methods like rotation, flipping, and scaling are also applied to increase dataset diversity and improve the model's ability to generalize on unseen data.

The system is developed using Convolutional Neural Networks (CNN), which is highly effective for image classification tasks. The CNN architecture consists of convolutional layers for feature extraction, pooling layers for reducing spatial dimensions, and fully connected layers for final classification. The convolutional layers automatically learn important image features such as shapes, edges, and textures from MRI scans, eliminating the need for manual feature extraction.

The model is trained using labeled MRI images with supervised learning techniques. The dataset is divided into training and testing sets to evaluate model performance accurately. The Adam optimizer and binary cross-entropy loss function are used during training to improve accuracy and minimize errors. After training, the model predicts whether a new MRI image contains a tumor or not. This automated methodology reduces manual effort and provides a reliable and efficient system for brain tumor detection.

4. Tools and Technologies

The proposed brain tumor detection system is developed using various modern tools and technologies related to Artificial Intelligence and Deep Learning. The programming language used for implementation is Python because of its simplicity and extensive support for machine learning and image processing libraries. The deep learning model is developed using TensorFlow and Keras frameworks, which provide powerful functionalities for building and training Convolutional Neural Networks (CNNs). These frameworks help in efficient model training, feature extraction, and image classification.

MRI image preprocessing and data handling are performed using libraries such as NumPy, OpenCV, and Matplotlib. NumPy is used for numerical operations and array handling, while OpenCV helps in image preprocessing tasks such as resizing and normalization. Matplotlib is used for visualizing graphs related to training accuracy, validation accuracy, and loss during model training. Data augmentation techniques such as rotation, flipping, and scaling are also applied to improve model performance and reduce overfitting.

The system is trained and tested using a publicly available MRI brain tumor dataset obtained from Kaggle. The development environment includes Jupyter Notebook or Google Colab, which provide an interactive platform for coding, model training, and result visualization. These tools and technologies together help in building an accurate, reliable, and efficient automated brain tumor detection system.

5. Implementation

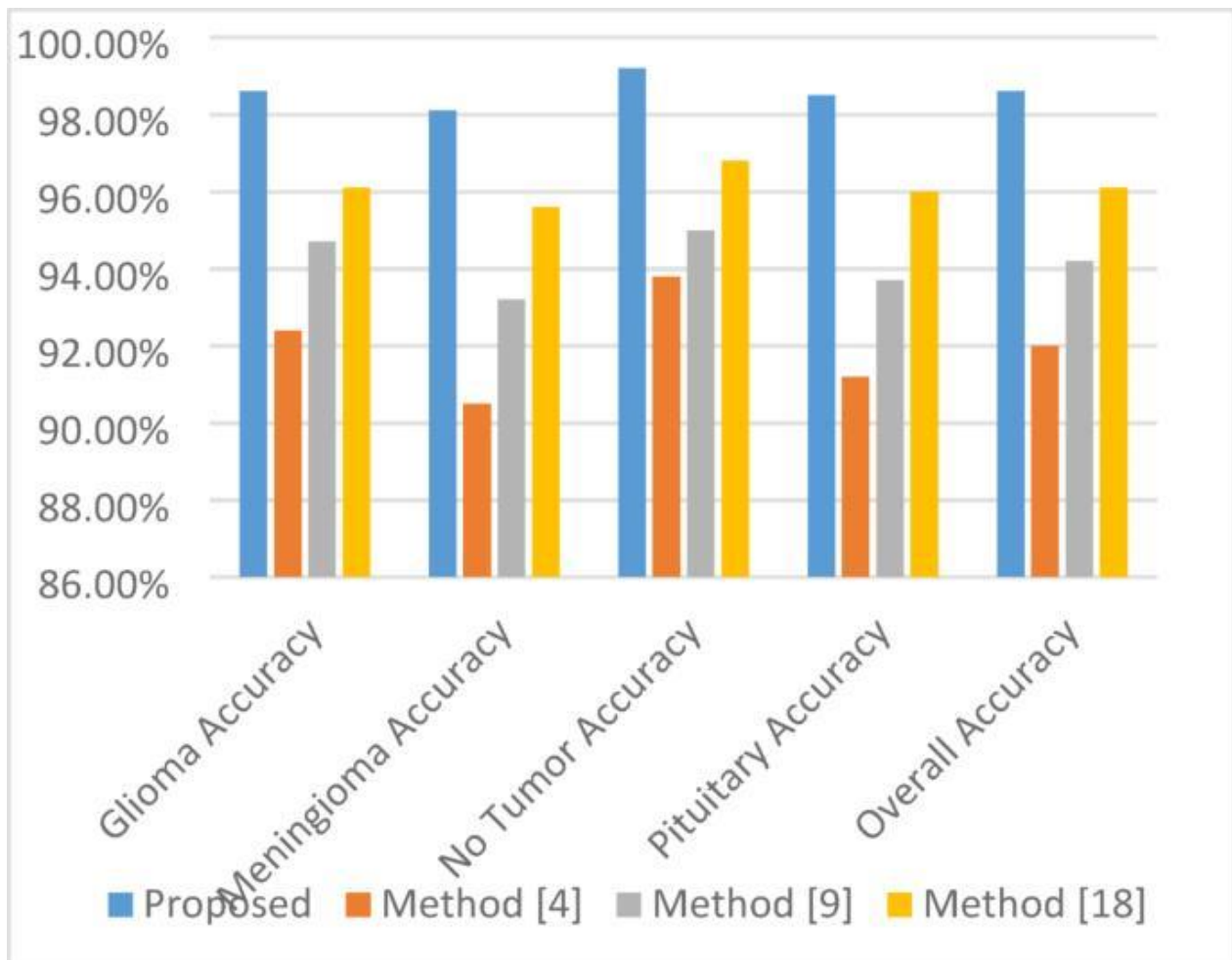
The implementation of the proposed brain tumor detection system is carried out using Python programming language along with deep learning frameworks such as TensorFlow and Keras. The entire system is developed in an interactive development environment like Jupyter Notebook or Google Colab. The MRI images are first loaded into the system and organized into tumor and non-tumor categories. Required libraries such as NumPy, OpenCV, Matplotlib, and TensorFlow are imported to perform image processing, visualization, and model training operations.

In the preprocessing stage, all MRI images are resized to a standard dimension of 224×224 pixels to maintain uniformity in the dataset. Pixel values are normalized to improve the training efficiency of the model. Data augmentation techniques including rotation, flipping, zooming, and scaling are applied to increase dataset diversity and reduce overfitting. After preprocessing, the dataset is divided into training and testing sets so that the model can be trained and evaluated properly.

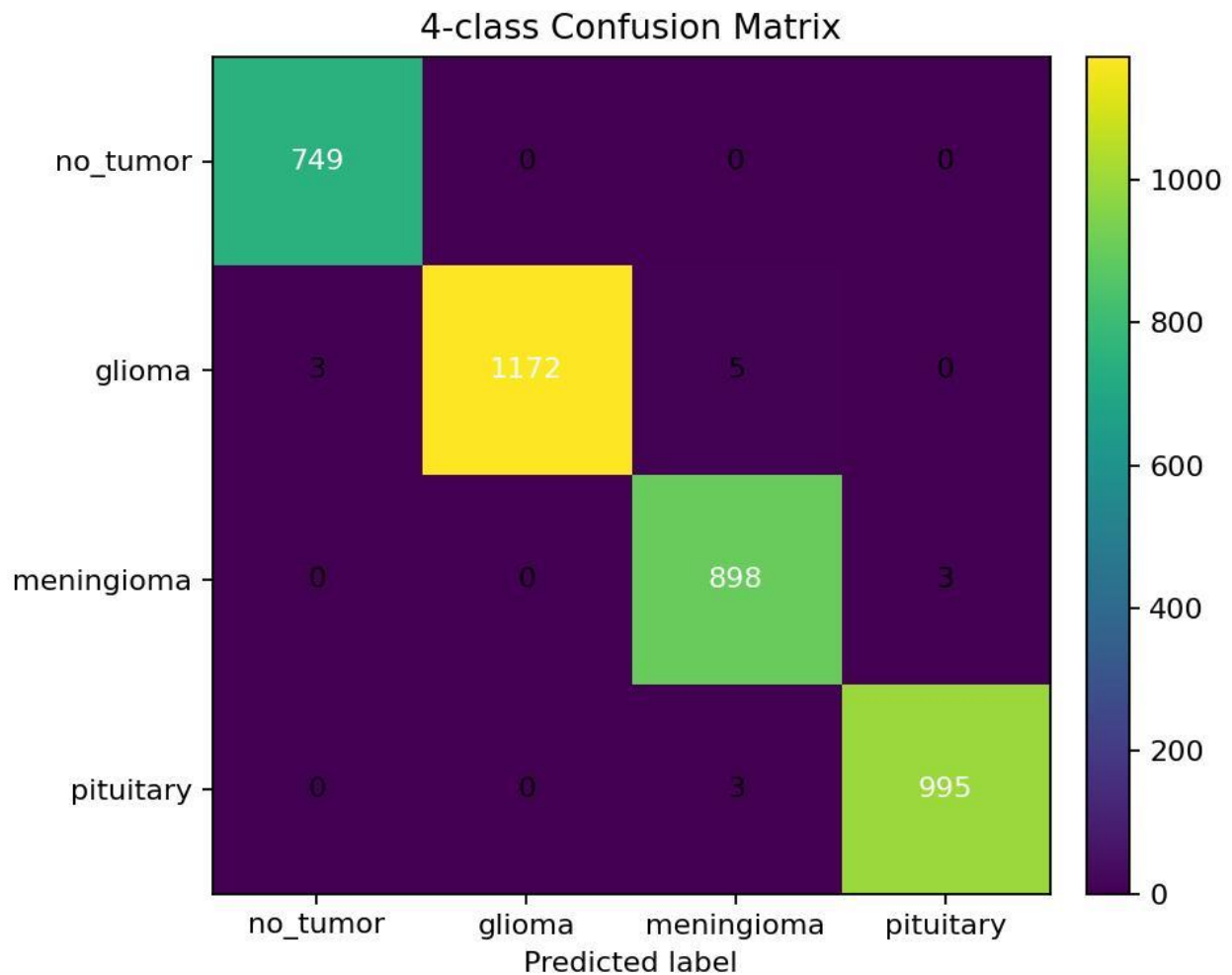
The proposed Convolutional Neural Networks (CNN) model is then implemented using multiple convolutional, pooling, and fully connected layers. The convolutional layers automatically extract important features from MRI images, while pooling layers reduce computational complexity. The model is trained using the Adam optimizer and binary cross-entropy loss function over several epochs with a fixed batch size. During training, the system continuously monitors accuracy and loss values to evaluate performance. After successful training, the model is tested using new MRI images to predict whether a brain tumor is present or not.

6. Major Findings

The proposed brain tumor detection system achieved high accuracy in classifying MRI images into tumor and non-tumor categories. The experimental results showed that the Convolutional Neural Networks (CNN) model successfully learned important features from MRI scans and performed better than many traditional machine learning approaches. The training and validation accuracy graphs indicated that the model converged effectively without significant overfitting, demonstrating the reliability of the deep learning approach for medical image classification tasks.



Another important finding of this research is that preprocessing techniques such as image resizing, normalization, and data augmentation significantly improved model performance. Data augmentation methods like rotation, flipping, and scaling increased dataset diversity and helped the system generalize better on unseen MRI images. The confusion matrix analysis also showed that the model correctly classified most tumor and non-tumor samples, proving the effectiveness of the proposed system in detecting brain abnormalities accurately.



The study also highlighted the potential of Deep Learning in healthcare and medical imaging applications. The automated detection system reduces the dependency on manual diagnosis, decreases human effort, and assists medical professionals in making faster and more accurate decisions. Although the current system focuses on binary classification, the findings suggest that future improvements using advanced architectures and larger datasets can further enhance the accuracy and expand the system for multi-class brain tumor classification.

7. Conclusion and Future Scope

This project presented a deep learning-based system for detecting brain tumors from MRI images using Convolutional Neural Networks (CNN). The proposed model successfully classified tumor and non-tumor images with high accuracy and reduced the need for manual diagnosis. Preprocessing techniques such as image resizing, normalization, and data augmentation further improved the performance and reliability of the system.

The study demonstrated that Deep Learning can play an important role in medical image analysis and assist doctors in making faster and more accurate decisions. The automated system reduces human effort and improves efficiency in detecting brain abnormalities from MRI scans.

In the future, the system can be improved by using advanced models such as ResNet and VGG16 to increase accuracy. The project can also be extended for multi-class classification to identify different types of brain tumors and integrated into real-time healthcare applications.

8. References

Link of Dataset:-

[https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor mri-dataset](https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset)

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